First ask whether the properties agree

Optional background page 1 | Use within the lesson when needed. No compulsory extra pre-reading.

Start the required route at week03-study.html. The video, six-page practice record and four-page guide supply the main route. This reading explains the concepts without adding another required session.

A fit and a thermodynamic constraint answer different questions

In Week 2, VLE measurements were used to infer γ_i and fit models. The low-pressure expression $\gamma_i = y_i P/(x_i P_i^{sat})$ assumes ideal vapor, negligible Poynting correction and applicable vapor-pressure data. Fitting residuals show reproduction under those assumptions. Consistency asks whether the inferred properties obey thermodynamic relationships.

One excess Gibbs function constrains two coefficients

For a binary liquid, $g = g^E/(RT) = x \ln \gamma_1 + (1-x) \ln \gamma_2$. Its composition derivative contains $\ln \gamma_1 - \ln \gamma_2$ plus $x\, d \ln \gamma_1/dx + (1-x)\, d \ln \gamma_2/dx$. At fixed T,P the latter sum vanishes by Gibbs-Duhem, giving $dg/dx = \ln(\gamma_1/\gamma_2)$. A local comparison tests this derivative relationship.

Along a general path, $\Sigma x_i\, d \ln \gamma_i = -h^E dT/(RT^2) + v^E dP/(RT)$. Isothermal VLE may vary in pressure; the teaching diagnostic neglects its excess-volume pressure term. Isobaric VLE varies in temperature and requires independent excess enthalpy for the corrected local assessment. Missing information does not make a test passed or failed.

A numerical derivative requires a stated procedure

The worksheet forms g at each supplied composition, differences g and subtracts the center log ratio. On a uniform grid, the centered slope is $(g_{right} - g_{left})/(x_{right} - x_{left})$. Lab 03 uses the corresponding three-point formula for actual spacing. Finite differences are approximations and may amplify noise. More sampled synthetic points do not create more independent experimental evidence.

C4/C5 are small fixed-T,P tables. Calculate them before using the Lab score. A single r is distinct from the dataset summary 100 mean |r|. Neither a one-point check nor three interior observations supplies a full-range area verdict.

Four questions to keep separate

Question	Evidence
Did the numerical solver converge?	Stopping/residual checks
Are thermodynamic constraints compatible?	Applicable diagnostics and assumptions
Does a model reproduce observations?	Fitting errors for that selected model
Does it predict independent data?	Evaluation on observations excluded from fitting

Global areas and local residuals

Optional background page 2 | Revisit after your Lab 03 predictions if you want the mechanism of the controlled defects.

Integrating is a useful reduction, but it loses location

The area method integrates $L=\ln(\gamma_1/\gamma_2)$ over the full composition range. The hand example uses $x=0,0.25,0.50,0.75,1$ and $L=2,1,0,-1,-2$. Its trapezoids sum to zero; adding 0.20 everywhere adds 0.20 to the signed integral. Analytic endpoint limits are supplied, rather than inferred from interior observations.

Lab 03 reports $D=100|A-B|/(A+B)$, with A/B the nonnegative areas above/below zero. This normalized difference is not the signed integral. It can be undefined near ideality. Extrapolating unknown endpoints supplies an assumption, not measurements.

A controlled defect separates the questions

Let $f=0.25\cos(2\pi x)$. Adding $(1-x)f$ to $\ln\gamma_1$ and $-xf$ to $\ln\gamma_2$ cancels in the weighted g but changes L by f. The full-range integral of f is zero while f is nonzero locally. The continuous local residual therefore changes by -f. The construction is deliberately inconsistent and demonstrates a diagnostic limitation, not a real chemical mixture.

A narrow localized defect provides a different challenge. A residual plot can show a peak while its average is small. Meeting an aggregate criterion does not imply that every local feature is negligible. In D4, compare 41 and 161 sampled points with all other settings fixed and report the actual change rather than assuming improvement.

Other approaches have their own scope

Herington: an empirical isobaric boiling-range allowance $J=150(T_{max}-T_{min})/T_{min}$, with kelvin temperatures and a D-J criterion. J is not measured h^E. It does not apply to the required isothermal runs.

Van Ness-type: test whether a chosen consistent model reproduces pressure and vapor composition. Lab 03 uses a teaching NRTL parameterization. Failure can reflect model flexibility, property approximations or data; close reproduction is not independent predictive validation.

Local diagnostic: finite differences of g, with a temperature correction where required. Lab 03 is not the fitted Padé/Kojima point-test implementation in NIST TDE. Its score and thresholds are teaching choices and should be reported with that scope.

A useful conclusion includes the settings, literal result/status, plot evidence and limitation. An overall “passed” label hides which physical and numerical questions were actually evaluated.

Primary descriptions: NIST Herington; NIST point test; NIST Van Ness. The Lab states its departures in Sources & scope.

Then ask which liquid state is stable

Optional background page 3 | Use after the consistency activities and the 95-105 minute break.

Consistency does not require one stable liquid

A Gibbs-Duhem-consistent model may have a nonconvex mixing-energy curve. At fixed T,P, equilibrium minimizes total Gibbs energy among admitted states satisfying material balances. Equal chemical potentials or solver convergence identifies a candidate; global stability still needs checking.

The common tangent between distinct compositions x^{α}/x^{β} has equal slopes and equal chemical potentials. It must support the whole admitted curve from below. Its segment gives the energy of realizable mixtures of the two endpoint liquids. Outside that segment, its extension is not a mixture with nonnegative amounts.

A homogeneous feed has a local and a global classification

Stable: no allowed split lowers its energy. **Metastable:** positive local curvature, yet a finite split lowers total energy. **Unstable:** negative curvature permits infinitesimal composition changes to lower energy. These labels describe the hypothetical homogeneous feed, not the final equilibrium phase count.

For symmetric Margules A=3, the investigation compares z_1=0.02, 0.12 and 0.50. Predict first, then inspect the computed energy curve. Use the actual calculated endpoints and phase fractions, not the illustrative hand values 0.20 and 0.80.

Balances and energies provide different evidence

$f\beta=(z_1-x^{\alpha})/(x^{\beta}-x^{\alpha})$, $f\alpha=1-f\beta$, $z_1=f\alpha x^{\alpha}+f\beta x^{\beta}$. Require $0\le f\alpha, f\beta\le 1$. A correct balance establishes feasibility, not the minimum energy. At z_1=0.02, extending the A=3 coexistence endpoints would require a negative amount and is inadmissible.

For this model, $m(x)=\Delta g_{mix}/(RT)=x \ln x+(1-x)\ln(1-x)+3x(1-x)$. This includes ideal mixing plus excess energy. Compare $m(z_1)$ with $f\alpha m(x^{\alpha})+f\beta m(x^{\beta})$. At a metastable feed, a finite energy decrease can coexist with positive local curvature.

Required evidence and optional extensions

Complete the three-feed record, two independent balances and the z_1=0.12 energy calculation. Annotate the supporting tangent and retain corrections to your predictions. Changing A explores different model behavior; it is an optional extension, not an additional required task. NRTL comparisons, LLE fitting and VLLE are also optional later work.

The model does not calculate nucleation barriers, interfacial energy, kinetics or domain patterns. A stable-equilibrium calculation does not predict how quickly separation occurs.

Original course explanation. Textbook reading locator from the supplied contents: sections 11.1-11.2, 14.11 and 16.2-16.4 for the stability bridge. No textbook text or figures are reproduced.